

Adithya Kishor

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B.Tech and MS in Computer Science and Engineering

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Github

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EDUCATION

Degree	Institute	CGPA	Year
B.Tech + M.S., CSE	International Institute of Information Technology, Hyderabad	8.47	2023-Present

EXPERIENCE

- Computer Systems Group, IIIT Hyderabad** *Jun. 2025 - Present*
Undergraduate Student Researcher Hyderabad
 - Design and implement performance-optimized code and algorithms across heterogeneous computer systems, including custom hardware accelerators (AMD Versal, Qualcomm, NPUs).
 - Apply ML-driven techniques to systems-level problems such as design space exploration, resource scheduling, and model quantization for on-device inference.
- MontyCloud** *May. 2025 - Jul. 2025*
Software Engineering Intern Bangalore
 - Built and optimized backend components for a multi-cloud SaaS platform, improving cloud resource adoption and cost-efficiency workflows.
- IIIT Hyderabad** *Jun. 2025 - Present*
Teaching Assistant Hyderabad
 - TA for Software Programming for Performance: mentored students on SIMD, OpenMP, and CUDA-based parallel programming and optimization.
 - TA for Embedded Systems Workshop: guided implementation and deployment of ML models on edge/Qualcomm hardware.
- Signal Processing and Communication Research Center, IIIT-H** *Dec. 2024 - Jan. 2025*
Research Intern Hyderabad
 - Deployed and quantized deep learning models (YOLO, Faster R-CNN, Llama, StableLM) for efficient on-device inference on Qualcomm hardware, including a RAG pipeline for Llama running natively on Android.

PROJECTS

- BitNet Inference Kernels on XDNA NPUs** *2025 - Present*
Academic Research
 - Designed and implemented optimized low-bit inference kernels for BitNet models on AMD XDNA NPUs, achieving 10 tok/s, outperforming RTX 4060 and matching modern CPU throughput at a fraction of the power draw; drafting a paper for publication.
- SWG Algorithm Acceleration on NPUs** *2025 - Present*
Academic Research
 - Engineered spatial-architecture kernels for the Smith-Waterman-Gotoh sequence alignment algorithm on XDNA NPUs, reaching 50 GCUPs by exploiting VLIW parallelism for high-throughput, low-power acceleration.
- Automated Course Leaderboard & Benchmarking System** *2025 - Present*
Teaching Assistant Project Github
 - Architected a multithreaded benchmarking queue service to concurrently evaluate 100+ student submissions in real time, powering a live class leaderboard used by an entire course cohort.
- Custom C Shell** *2025 - Present*
Operating Systems Course Github
 - Built a feature-rich Unix shell in C supporting I/O redirection, piping, background process management, aliasing, and signal handling; wrote custom system-inspection commands.
- xv6 Schedulers & Copy-on-Write Memory Management** *2025 - Present*
Operating Systems Course Github
 - Implemented Round Robin, MLFQ, and Lottery scheduling algorithms and Copy-on-Write page fault handling in the xv6 RISC-V kernel to reduce memory allocation overhead on process creation.

SKILLS

- Languages:** Python, C/C++, Java, JavaScript, Rust, Kotlin
- Systems & Backend:** Multithreading/Concurrency, OS Internals, SIMD/OpenMP/CUDA, REST APIs, Distributed Systems, Cloud (AWS), Bash
- Tools & Frameworks:** Git, PyTorch, TensorFlow, Flask, MySQL, Docker

ACHIEVEMENTS & LEADERSHIP

- Co-Founder, Coconut Phishers**, CTF team, top 15 in India & sub-300 globally; co-organized The Deccan CTF *Dec. 2023 - Present*
- Winner & Runner-Up, Megathon**, Qualcomm Challenge, working with Snapdragon NPUs *2024, 2025*
- Best Framework Contribution**, HackIIIT - MOYA Challenge: extended MOYA, a custom multi-agent orchestration framework, with native tool-calling and vectorstore-backed memory for autonomous agent workflows *2025*